QBS 103 Final Project - Submission 1

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Data Import

```
# Import the original datasets
covar <- read.csv("GSE85307_covar-1.csv")

gene_expr <- read.csv(
  "GSE85307_geneExpression.csv",
  row.names = 1,
  check.names = FALSE
)

gene_names <- read.csv("GSE85307_geneNames.csv")
```

Initial Data Inspection

```
# Check the dimensions of the three datasets
dim(covar)
```

```
## [1] 157  10
```

```
dim(gene_expr)
```

```
## [1] 500 157
```

```
dim(gene_names)
```

```
## [1] 750   2
```

```
# Inspect variable names and sample identifiers
names(covar)
```

```
##  [1] "subjectID"                  "geoID"
##  [3] "mat_age_yrs"                "preg_preeclampsia"
##  [5] "delivery_gestAge_wks"       "sampleCollection_gestAge_wks"
##  [7] "batch_Microarray"           "mat_BMI"
##  [9] "mat_vitD"                   "mat_asthma"
```

```r
head(colnames(gene_expr))
```

```
## [1] "GSM2264363" "GSM2264379" "GSM2264394" "GSM2264418" "GSM2264449"
## [6] "GSM2264452"
```

```r
head(rownames(gene_expr))
```

```
## [1] "8089659" "8083594" "8035050" "8141107" "8093624" "7969472"
```

```r
names(gene_names)
```

```
## [1] "PROBEID" "GENE"
```

Sample ID Validation

```r
# Check whether all covariate sample IDs exist in the expression matrix
all(covar$geoID %in% colnames(gene_expr))
```

```
## [1] TRUE
```

```r
# Check whether all expression sample IDs exist in the covariate data
all(colnames(gene_expr) %in% covar$geoID)
```

```
## [1] TRUE
```

```r
# Check for duplicated sample IDs in both datasets
anyDuplicated(covar$geoID)
```

```
## [1] 0
```

```r
anyDuplicated(colnames(gene_expr))
```

```
## [1] 0
```

```r
# Check whether the sample IDs are in the same order
identical(covar$geoID, colnames(gene_expr))
```

```
## [1] FALSE
```

Sample Alignment

```
# Identify the positions of covariate sample IDs in the expression matrix
sample_order <- match(covar$geoID, colnames(gene_expr))

# Reorder expression columns to match the covariate sample order
gene_expr_aligned <- gene_expr[, sample_order, drop = FALSE]

# Verify that the sample IDs are correctly aligned
identical(covar$geoID, colnames(gene_expr_aligned))
```

```
## [1] TRUE
```

Covariate Inspection

```
# Inspect the original covariate data structure
str(covar)
```

```
## 'data.frame':    157 obs. of  10 variables:
##  $ subjectID                   : chr  "6053887-M" "3064425-M" "6094898-M" "7050617-M" ...
##  $ geoID                       : chr  "GSM2264346" "GSM2264347" "GSM2264348" "GSM2264349" ...
##  $ mat_age_yrs                 : chr  "age_years: 31.06639288" "age_years: 27.79466119" "age_years: 3
##  $ preg_preeclampsia           : chr  "pregnancy_condition: Preeclampsia" "pregnancy_condition: Norma
##  $ delivery_gestAge_wks        : chr  "gestation_weeks: 37.42857143" "gestation_weeks: 39.42857143"
##  $ sampleCollection_gestAge_wks: chr  "blood_draw_at_gestation_week: 10" "blood_draw_at_gestation_we
##  $ batch_Microarray            : chr  "microarray_hybridization_date: 03/15/2013" "microarray_hybridi
##  $ mat_BMI                     : chr  "bmi_kg/m2: 37.52407538" "bmi_kg/m2: 20.59144273" "bmi_kg/m2: 1
##  $ mat_vitD                    : chr  "vitamind_25(0h)_ng/ml_whole_blood_baseline: 7.94" "vitamind_25
##  $ mat_asthma                  : chr  "asthma_0=healthy_1=asthmatic: 1" "asthma_0=healthy_1=asthmatic
```

```
# Check the data types of all covariates
sapply(covar, class)
```

```
##                    subjectID                        geoID
##                  "character"                  "character"
##                  mat_age_yrs              preg_preeclampsia
##                  "character"                  "character"
##         delivery_gestAge_wks sampleCollection_gestAge_wks
##                  "character"                  "character"
##             batch_Microarray                      mat_BMI
##                  "character"                  "character"
##                     mat_vitD                   mat_asthma
##                  "character"                  "character"
```

```
# Check for missing values in the original data
colSums(is.na(covar))
```

```
##                    subjectID                        geoID
##                            0                            0
```

```
##                  mat_age_yrs            preg_preeclampsia
##                            0                            0
##         delivery_gestAge_wks sampleCollection_gestAge_wks
##                            0                            0
##             batch_Microarray                      mat_BMI
##                            0                            0
##                      mat_vitD                   mat_asthma
##                            0                            0
```

Data Cleaning

```
# Create a working copy and clean the continuous variables
covar_clean <- covar %>%
  mutate(
    # Remove text prefixes and convert maternal age and BMI to numeric
    mat_age_yrs = as.numeric(str_remove(mat_age_yrs, ".*: ")),
    mat_BMI = as.numeric(str_remove(mat_BMI, ".*: "))
  )
```

Data Cleaning Validation

```
# Inspect the data structure after cleaning
glimpse(covar_clean)
```

```
## Rows: 157
## Columns: 10
## $ subjectID                   <chr> "6053887-M", "3064425-M", "6094898-M", "7~
## $ geoID                       <chr> "GSM2264346", "GSM2264347", "GSM2264348",~
## $ mat_age_yrs                 <dbl> 31.06639, 27.79466, 34.31896, 33.42642, 2~
## $ preg_preeclampsia           <chr> "pregnancy_condition: Preeclampsia", "pre~
## $ delivery_gestAge_wks        <chr> "gestation_weeks: 37.42857143", "gestatio~
## $ sampleCollection_gestAge_wks <chr> "blood_draw_at_gestation_week: 10", "bloo~
## $ batch_Microarray            <chr> "microarray_hybridization_date: 03/15/201~
## $ mat_BMI                     <dbl> 37.52408, 20.59144, 20.71119, 31.99358, 2~
## $ mat_vitD                    <chr> "vitamind_25(0h)_ng/ml_whole_blood_baseli~
## $ mat_asthma                  <chr> "asthma_0=healthy_1=asthmatic: 1", "asthm~
```

```
# Check for missing values introduced during conversion
sum(is.na(covar_clean$mat_age_yrs))
```

```
## [1] 0
```

```
sum(is.na(covar_clean$mat_BMI))
```

```
## [1] 0
```

```r
# Review the distributions of the cleaned variables
summary(covar_clean$mat_age_yrs)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   18.43   22.15   26.57   26.38   30.14   38.76
```

```r
summary(covar_clean$mat_BMI)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   17.95   22.90   27.33   28.94   33.27   61.95
```

Categorical Variable Inspection

```r
# Check the original categories of pregnancy outcome
table(covar_clean$preg_preeclampsia, useNA = "ifany")
```

```
##
##       pregnancy_condition: Normal pregnancy_condition: Preeclampsia
##                               110                               47
```

```r
# Check the number of observations below and above BMI 25
table(covar_clean$mat_BMI >= 25, useNA = "ifany")
```

```
##
## FALSE  TRUE
##    53   104
```

Categorical Variable Cleaning

```r
# Clean pregnancy outcome and create BMI categories
covar_clean <- covar_clean %>%
  mutate(
    # Remove the text prefix from pregnancy outcome
    preg_preeclampsia = str_remove(preg_preeclampsia, ".*: "),

    # Convert pregnancy outcome to a categorical variable
    preg_preeclampsia = factor(
      preg_preeclampsia,
      levels = c("Normal", "Preeclampsia")
    ),

    # Categorize maternal BMI using the cutoff of 25 kg/m2
    BMI_group = case_when(
      mat_BMI < 25 ~ "BMI < 25",
      mat_BMI >= 25 ~ "BMI >= 25",
      TRUE ~ NA_character_
    ),
```

```
    # Set the order of BMI categories
    BMI_group = factor(
      BMI_group,
      levels = c("BMI < 25", "BMI >= 25")
    )
  )
# Check the number of observations in each combination
count(covar_clean, preg_preeclampsia, BMI_group)
```

```
##   preg_preeclampsia BMI_group  n
## 1            Normal  BMI < 25 40
## 2            Normal BMI >= 25 70
## 3      Preeclampsia  BMI < 25 13
## 4      Preeclampsia BMI >= 25 34
```

```
# Check for missing values after categorization
colSums(is.na(covar_clean[, c("preg_preeclampsia", "BMI_group")]))
```

```
## preg_preeclampsia         BMI_group
##                 0                 0
```

Gene Selection

```
# Inspect the structure of the gene annotation dataset
glimpse(gene_names)
```

```
## Rows: 750
## Columns: 2
## $ PROBEID <int> 8128043, 7962489, 7920271, 7900468, 8026496, 7989725, 8102362,~
## $ GENE    <chr> "CNR1", "PLEKHA8P1", "S100A4", "NFYC", "LINC00661", "PDCD7", "~
```

```
# Find the probe IDs associated with PTX3
gene_names %>%
  filter(GENE == "PTX3")
```

```
##   PROBEID GENE
## 1 8083594 PTX3
```

```
# Check whether the corresponding probes exist in the expression matrix
gene_names %>%
  filter(GENE == "PTX3") %>%
  mutate(in_expression_data = as.character(PROBEID) %in%
           rownames(gene_expr_aligned))
```

```
##   PROBEID GENE in_expression_data
## 1 8083594 PTX3               TRUE
```

Analysis Dataset

```
# Extract PTX3 expression values from the aligned expression matrix
ptx3_expression <- as.numeric(
  gene_expr_aligned["8083594", ]
)

# Add PTX3 expression to the cleaned covariate dataset
analysis_data <- covar_clean %>%
  mutate(PTX3_expression = ptx3_expression)

# Check the dimensions of the analysis dataset
dim(analysis_data)
```

```
## [1] 157  12
```

```
# Inspect the selected variables
analysis_data %>%
  select(geoID, mat_age_yrs, mat_BMI,
         preg_preeclampsia, BMI_group, PTX3_expression) %>%
  head()
```

```
##        geoID mat_age_yrs  mat_BMI preg_preeclampsia BMI_group PTX3_expression
## 1 GSM2264346    31.06639 37.52408      Preeclampsia BMI >= 25        4.674865
## 2 GSM2264347    27.79466 20.59144            Normal  BMI < 25        4.670370
## 3 GSM2264348    34.31896 20.71119            Normal  BMI < 25        4.793649
## 4 GSM2264349    33.42642 31.99358            Normal BMI >= 25        5.142027
## 5 GSM2264350    28.27926 27.17734      Preeclampsia BMI >= 25        4.843466
## 6 GSM2264351    23.50171 22.28685            Normal  BMI < 25        4.737083
```

```
# Check for missing gene expression values
sum(is.na(analysis_data$PTX3_expression))
```

```
## [1] 0
```

```
# Summarize PTX3 expression levels
summary(analysis_data$PTX3_expression)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   4.552   4.776   4.867   4.884   4.951   5.651
```

Data Visualization

Figure 1. Distribution of PTX3 Gene Expression

```
# Plot the distribution of PTX3 gene expression
ggplot(analysis_data, aes(x = PTX3_expression)) +
  geom_histogram(
```

```
    bins = 15,
    fill = "blue",
    color = "white",
    linewidth = 0.4
  ) +
  labs(
    title = "Distribution of PTX3 Gene Expression",
    x = "PTX3 Expression Level",
    y = "Number of Samples"
  ) +
  theme_classic(base_size = 12)
```

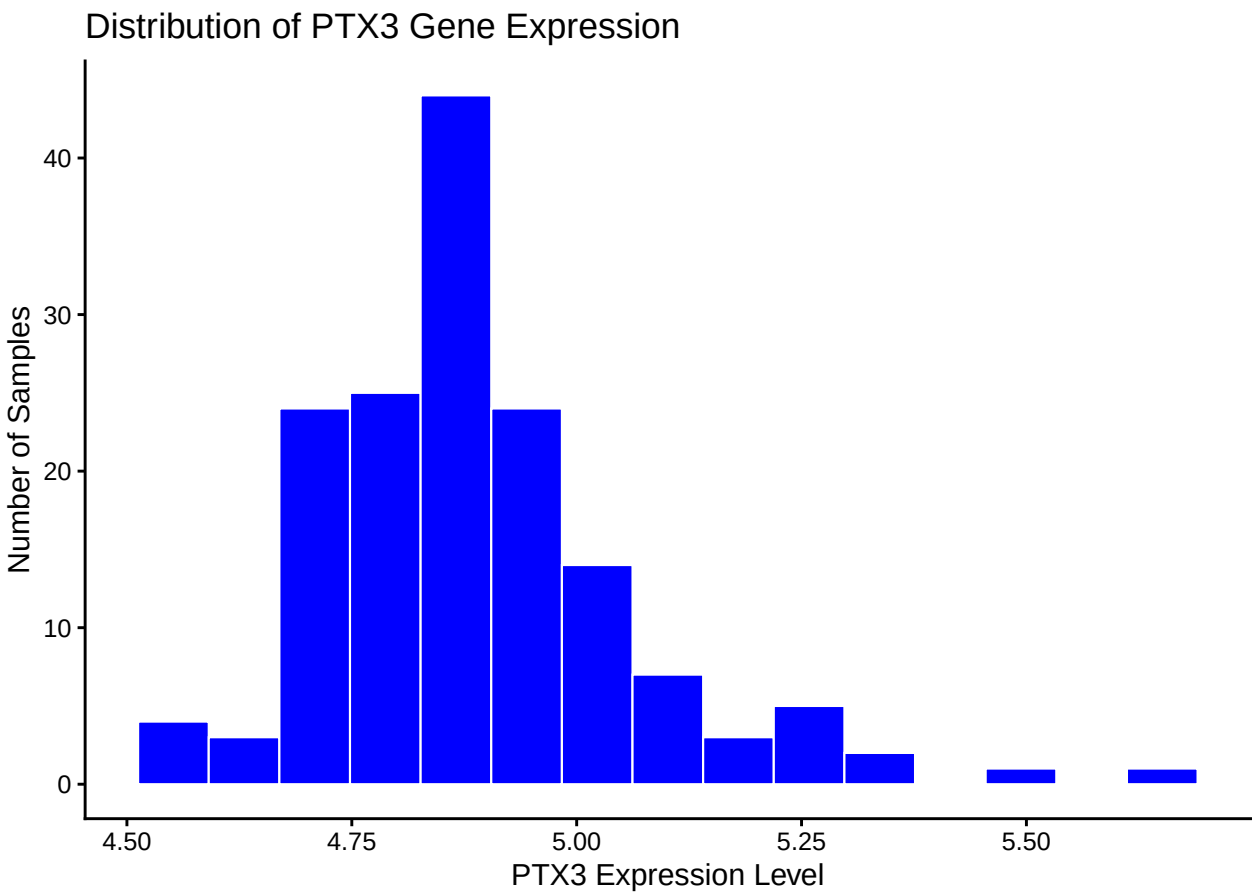


Figure 2. Maternal Age and PTX3 Expression

```
# Examine the relationship between maternal age and PTX3 expression
ggplot(analysis_data, aes(x = mat_age_yrs, y = PTX3_expression)) +
  geom_point(
    color = "blue",
    alpha = 0.7,
    size = 2
  ) +
  geom_smooth(
    method = "lm",
```

```
    se = TRUE,
    color = "brown",
    linewidth = 0.8
  ) +
  labs(
    x = "Maternal Age (years)",
    y = "PTX3 Expression Level"
  ) +
  theme_classic(base_size = 12)
```

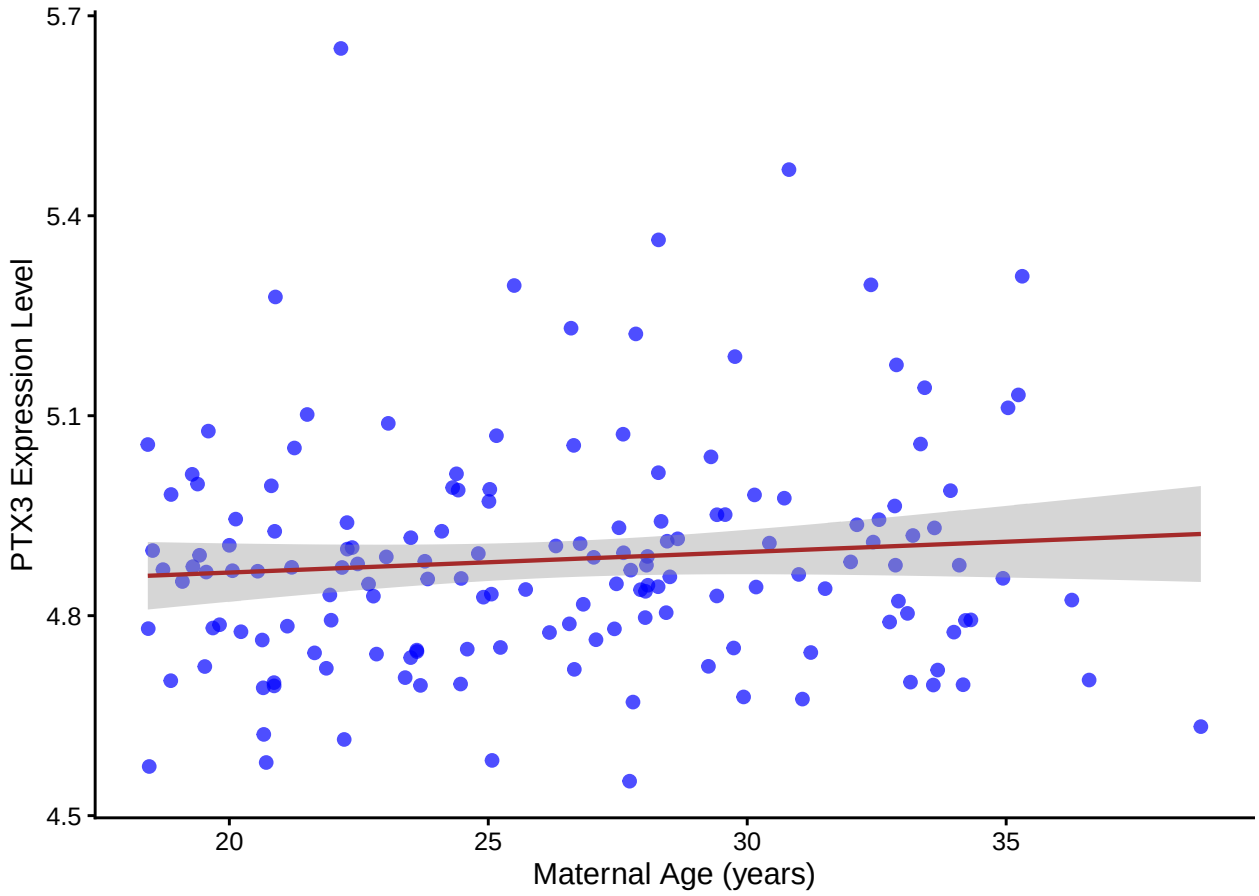


Figure 1: Figure 2. Relationship between maternal age and PTX3 gene expression.

Figure 3. PTX3 Expression by Pregnancy Outcome and BMI Group

```
# Compare PTX3 expression across pregnancy outcomes and BMI groups
ggplot(
  analysis_data,
  aes(x = preg_preeclampsia, y = PTX3_expression, fill = BMI_group)
) +
  geom_boxplot(
    position = position_dodge(width = 0.8),
    width = 0.65,
```

```
    alpha = 0.8,
    outlier.shape = 16,
    outlier.size = 1.5
  ) +
  scale_fill_manual(
    values = c("BMI < 25" = "blue",
               "BMI >= 25" = "brown")
  ) +
  labs(
    x = "Pregnancy Outcome",
    y = "PTX3 Expression Level",
    fill = "Maternal BMI"
  ) +
  theme_classic(base_size = 12) +
  theme(
    legend.position = "top"
  )
```

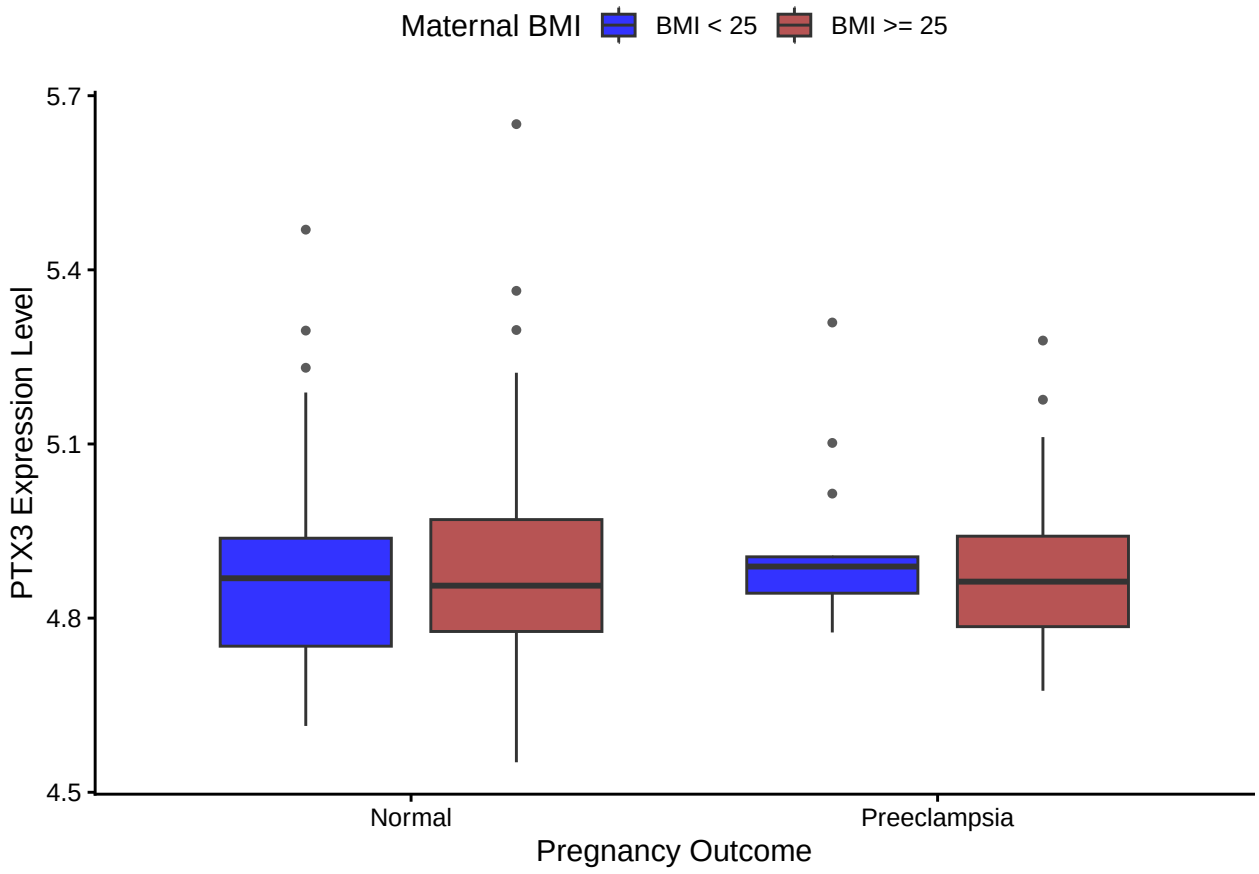


Figure 2: Figure 3. PTX3 gene expression by pregnancy outcome and maternal BMI group.